

Brayden Noh

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www.braydennoh.github.io

Education

Harvard University

Cambridge, MA

- Ph.D. in Earth and Planetary Sciences (Geophysics)

Expected 2030

California Institute of Technology (Caltech)

Pasadena, CA

- B.S. in Geological and Planetary Sciences

2025

Experiences

Harvard Meade Group, *Graduate Student*

Sep. 2025 – Present

- Building qsDEM, a quasi-static discrete-element model that couples brittle lithospheric faulting with viscous mantle flow to simulate subduction, mountain building, and planetary deformation.
- Developed a kinematic model that unifies long-term uplift with the short-term earthquake cycle.

Caltech Lamb Surface Processes Group, *Undergraduate Student*

Dec 2021 – Jul 2025

- Built Bayesian models of channel evolution in meandering rivers, producing probabilistic hazard forecasts.
- Developed image-based particle tracking algorithm to quantify fine sediment (mud) transport.

Publications

- Noh, B. A percolation transition for Scheidegger river networks, *Physica A: Statistical Mechanics and its Applications*
- Noh, B. and Wani, O. Cutoffs as a sufficient condition for chaos in kinematic river channel evolution, *Nature Communications Earth & Environment*
- Noh, B., Nghiem, J. A., Miller, K. L., Wang, D., & Lamb, M. P. Constraining the distribution of 3D fractal structures in mud flocs, *Journal of Geophysical Research: Earth Surface*
- Noh, B., Wani, O., Dunne, K. B. J., & Lamb, M. P. Geomorphic risk maps for river migration using probabilistic modeling – a framework, *Earth Surface Dynamics*
- Noh, B. Development of a supercapacitor hybrid powertrain design with pulse-width modulation and series configuration for light electric vehicles, *SAE International Journal of Electrified Vehicles*
- Noh, B. Fabrication and empirical analysis of graphene dispersion/activated carbon on conductive networks in porous graphite felt supercapacitor, *Journal of Energy Storage*

Awards and Grants

- 2025 NSF Graduate Research Fellowship Program (GRFP)
- 2025 Caltech Joel & Marcella Bonsall SURF Technical Writing Prize (3rd place)
- 2025 Caltech Doris S. Perpall SURF Speaking Competition Semi-Finalist
- 2024 ThinkSwiss Summer School Travel Grant: Awarded by the Embassy of Switzerland to attend the Swiss Federal Institute of Aquatic Science and Technology (EAWAG) Summer School in Environmental System Analysis
- 2024 Caltech James Michelin Scholarship: Recognizes an undergraduate student for contributions to geophysics
- 2023 Caltech Howard Reynolds Memorial Prize: Awarded to a student who demonstrates the potential to excel in the field of geology and who actively contributes to the quality of Caltech student life
- 2021 AGU Michael H. Freilich Student Visualization: Awarded the Grand Prize, which includes a travel grant and presentation on the NASA Hyperwall at the 2021 AGU Fall Meeting
- 2018, 2019, 2020, 2021 International Science and Engineering Fair (ISEF) Finalist: Awarded the Grand Prize in Energy in 2019